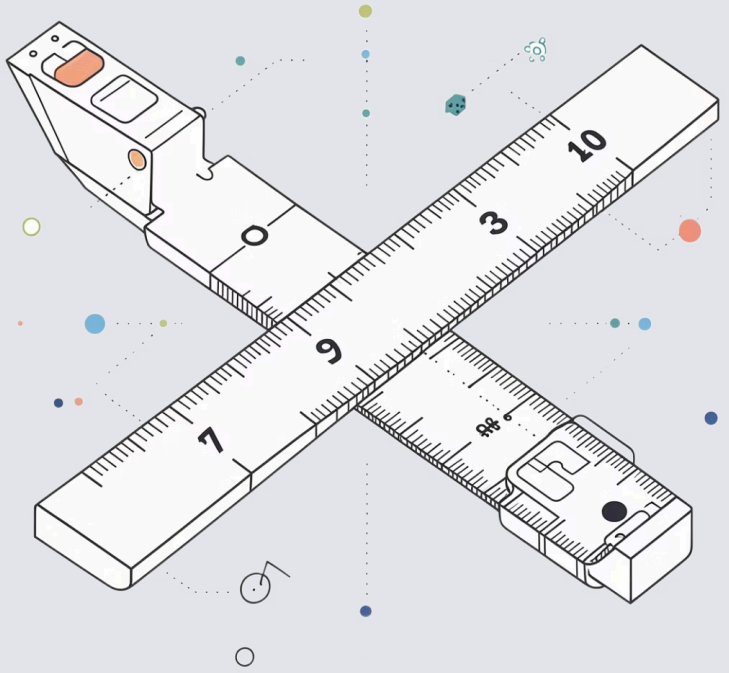
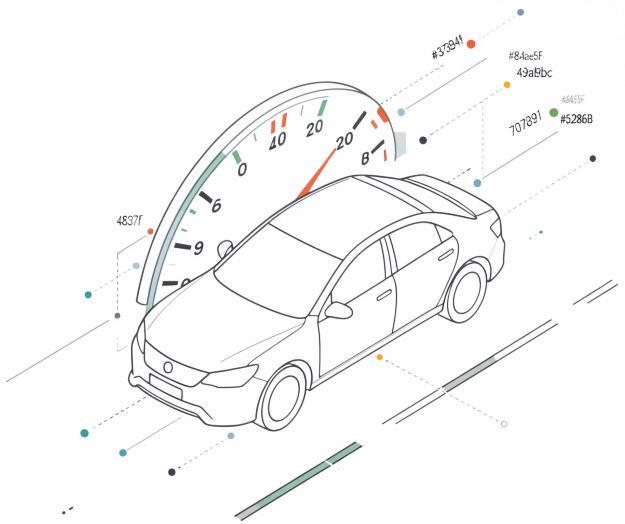
SI Units & Unit Conversion

Class 11 Physics · 30-Minute Lecture



Made with GAMMA



HOOK QUESTION

Why can't we simply say the speed of a car is 20?

What's missing?

Think about it — 20 what? Miles? Kilometers? Meters per second?

A number alone is meaningless without a unit.

Made with GAMMA



What is a Measurement?

Number

The magnitude — how much

Unit

The standard — of what

Together

5 meters · 10 seconds · 2 kilograms

Made with GAMMA

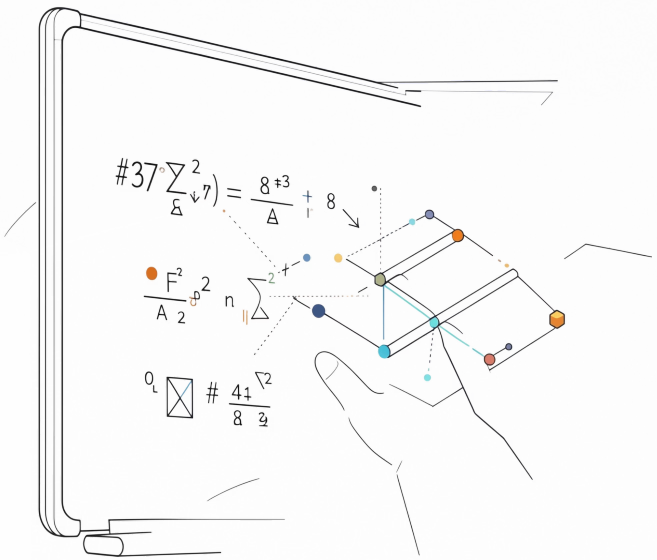
The 7 SI Base Units

Quantity	Unit	Symbol
Length	Meter	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	K
Electric Current	Ampere	A
Luminous Intensity	Candela	cd
Amount of Substance	Mole	mol

Made with GAMMA

Derived Units

Definition: Units formed by combining SI base units through multiplication or division.



Force – Newton (N)

$kg \cdot m/s^2$

Energy – Joule (J)

$kg \cdot m^2/s^2$

Power – Watt (W)

$kg \cdot m^2/s^3$

Made with GAMMA

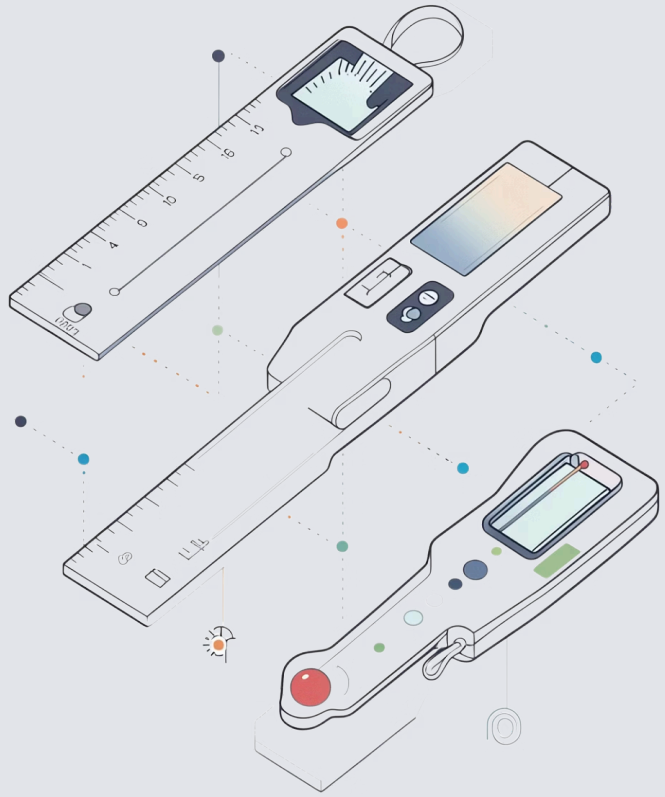
Why Convert Units?

The Problem

Real-world data arrives in mixed, inconsistent units

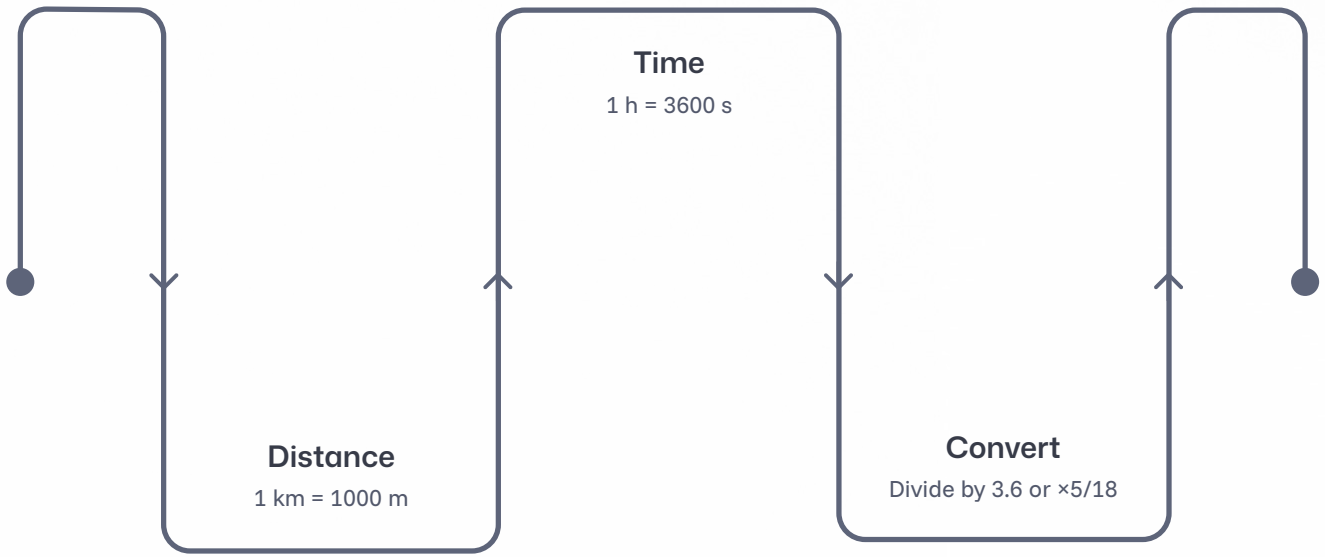
The Requirement

Physics equations demand complete unit consistency on both sides



Made with GAMMA

Converting km/h → m/s



DERIVE ON BOARD

Key Relation

1 km/h = 5/18 m/s ≈ 0.278 m/s

Notes

Follow the derivation on the board step by step.

Made with GAMMA

DERIVE ON BOARD

Converting m/s → km/h

Reciprocal of the previous conversion

Key Relation

1 m/s = 18/5 km/h = 3.6 km/h

Multiply by 3.6 to convert from m/s to km/h.

Quick Reference

→ **km/h to m/s:** × 5/18

→ **m/s to km/h:** × 18/5

→ **m/s to km/h:** × 3.6

My Notes

Made with GAMMA

Practice: Speed Conversions

1

72 km/h = ? m/s

72 × (5/18) = **20 m/s**

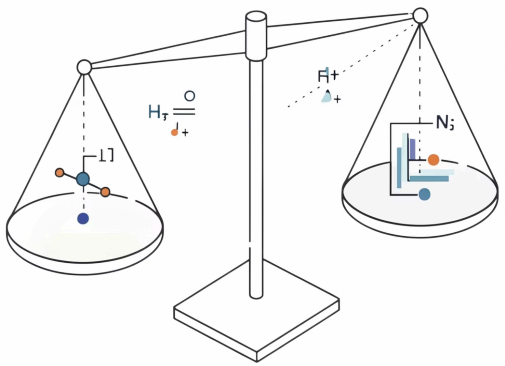
2

10 m/s = ? km/h

10 × (18/5) = **36 km/h**

Space for your working below ↓

Made with GAMMA



Dimensional Analysis

Rule: Both sides of any equation must carry the same dimensions.

Dimensions don't lie — use them to verify every formula.

Check F = ma

$[\text{kg} \cdot \text{m/s}^2] = [\text{kg}] \times [\text{m/s}^2]$ ✓

Notes

Made with GAMMA

Exit Question

> *Why can't we simply say the speed of a car is 20?*

Write your answer in the space below — in one complete sentence, explain what makes a measurement meaningful.



Made with GAMMA